NLP AND CV FINAL PROJECT

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This project presents a courtroom scene analyzer that integrates computer vision and natural language processing to analyze both visual and audio data from courtroom videos. A YOLO-based model is used to detect key objects such as individuals, the judge's bench, flags, and microphones, while OpenAI's Whisper model is used to transcribe spoken dialogue.

The transcribed text is then processed to generate summaries, extract keywords, and support a transcript-based chatbot capable of answering questions about the video. The results demonstrate how combining visual detection with language understanding can provide a more complete interpretation of real-world environments such as legal proceedings.

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INTRODUCTION

1. This project develops a Courtroom Scene Analyzer using Computer Vision (CV), and Natural Language Processing (NLP). The Courtroom Scene Analyzer will utilize CV for detecting and tracking specific items in the courtroom; NLP will be utilized to capture spoken argument statements from court proceedings. Each component of the system has been developed as independent models before being combined.

2. Computer Vision Component

2.1 Overview

The computer vision portion of this project uses the YOLOv8 model to detect objects within courtroom scenes. The goal is to identify relevant elements such as people, the judge's bench, flags, and microphones.

2.2 Dataset Creation

A custom dataset was created for this project. Images were collected from courtroom videos and manually filtered to ensure relevance and clarity. The dataset was labeled using RoboFlow, where bounding boxes were drawn for each object. The dataset consisted of approximately 63 images, split into training and validation sets. Images were extracted from courtroom videos and manually filtered before labeling.

The selected classes include:

- Person
- Judge Bench
- Flag
- Microphone

All images were reviewed and adjusted to maintain consistent labeling across the dataset.

2.3 Training Process

The YOLO model was trained using the labeled dataset. Training parameters were selected to balance detection accuracy and performance.

Key parameters:

- Confidence threshold: 0.5
- IoU threshold: 0.4
- Multiple training epochs

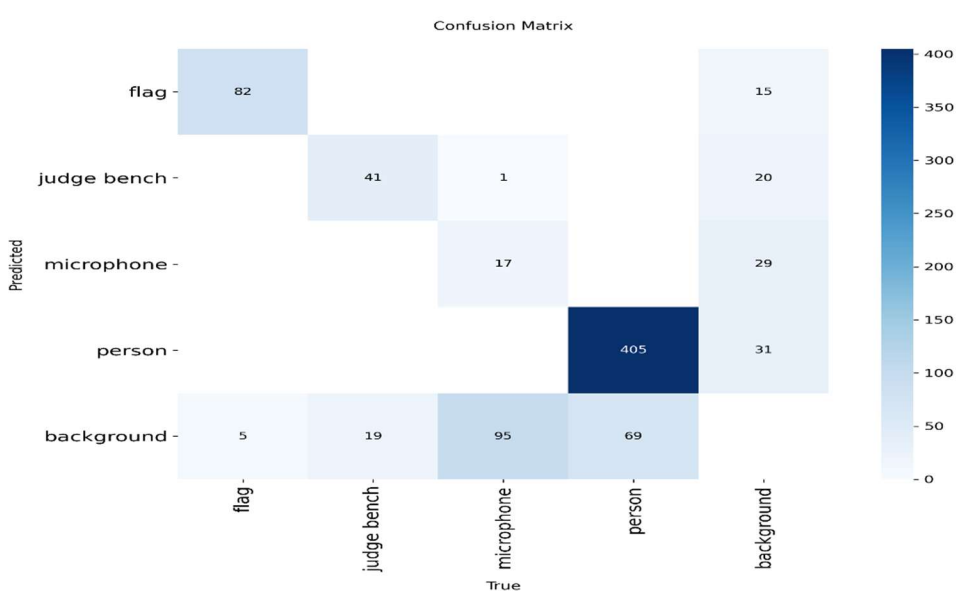
Training Results:

```
!ls runs/detect/train

args.yaml                       results.csv         val_batch0_labels.jpg
BoxF1_curve.png                 results.png         val_batch0_pred.jpg
BoxP_curve.png                  train_batch0.jpg    val_batch1_labels.jpg
BoxPR_curve.png                 train_batch160.jpg  val_batch1_pred.jpg
BoxR_curve.png                  train_batch161.jpg  val_batch2_labels.jpg
confusion_matrix_normalized.png train_batch162.jpg  val_batch2_pred.jpg
confusion_matrix.png            train_batch1.jpg    weights
labels.jpg                      train_batch2.jpg
```

This graph shows how the model improved over time, including reductions in loss and increases in accuracy.

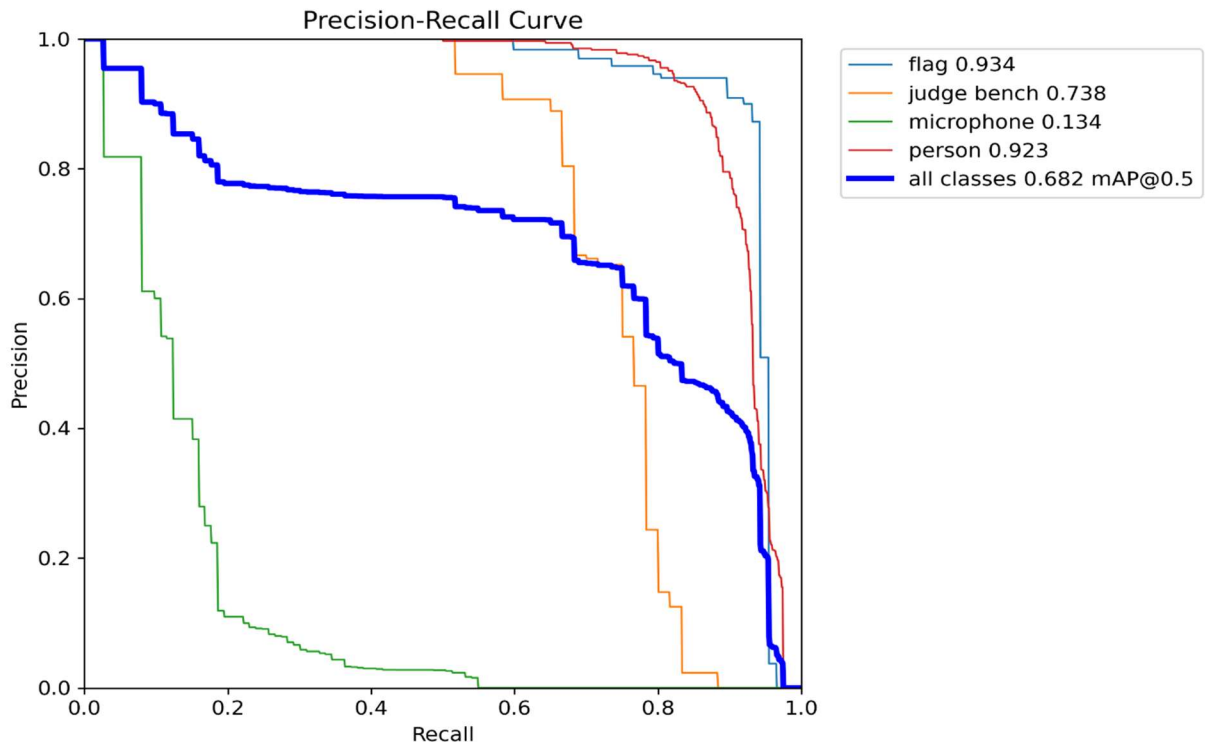
Confusion Matrix :



The confusion matrix shows how accurately the model classifies each object and highlights any misclassifications.

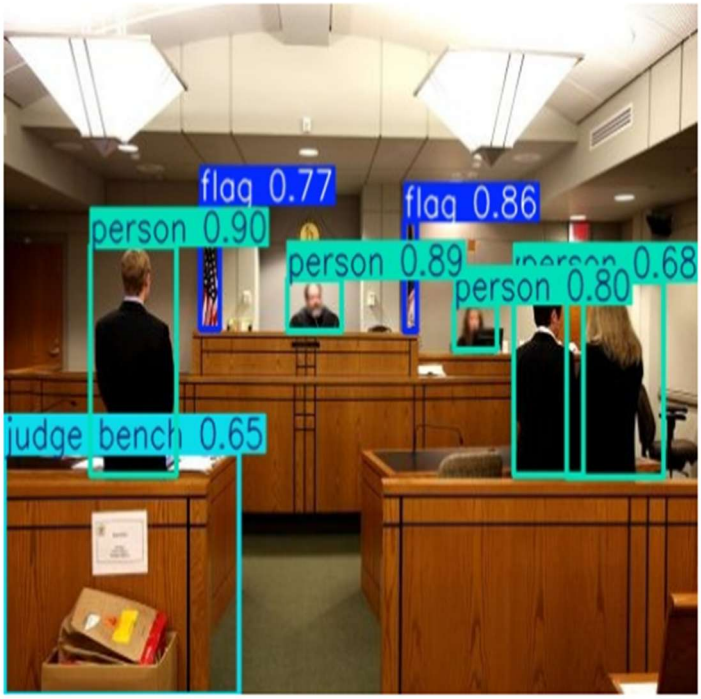
This curve shows the trade-off between precision and recall for the model. It shows the relationship between precision and recall for each class detected by the model. Precision measures how accurate the predictions are, while recall measures how many actual objects were correctly detected

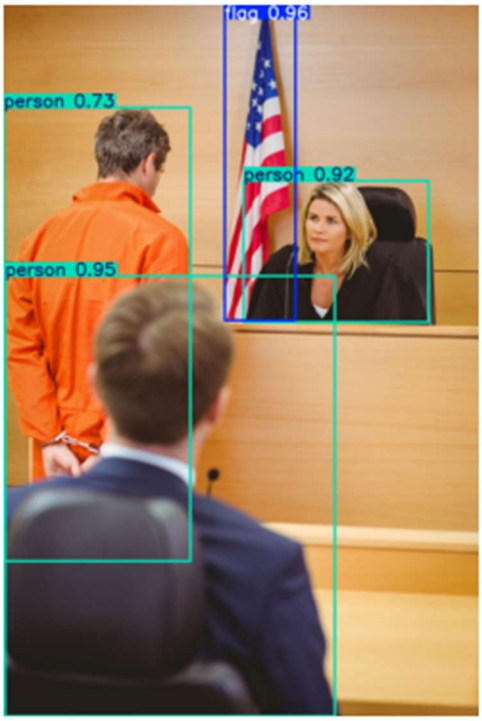
Precision-Recall Curve:



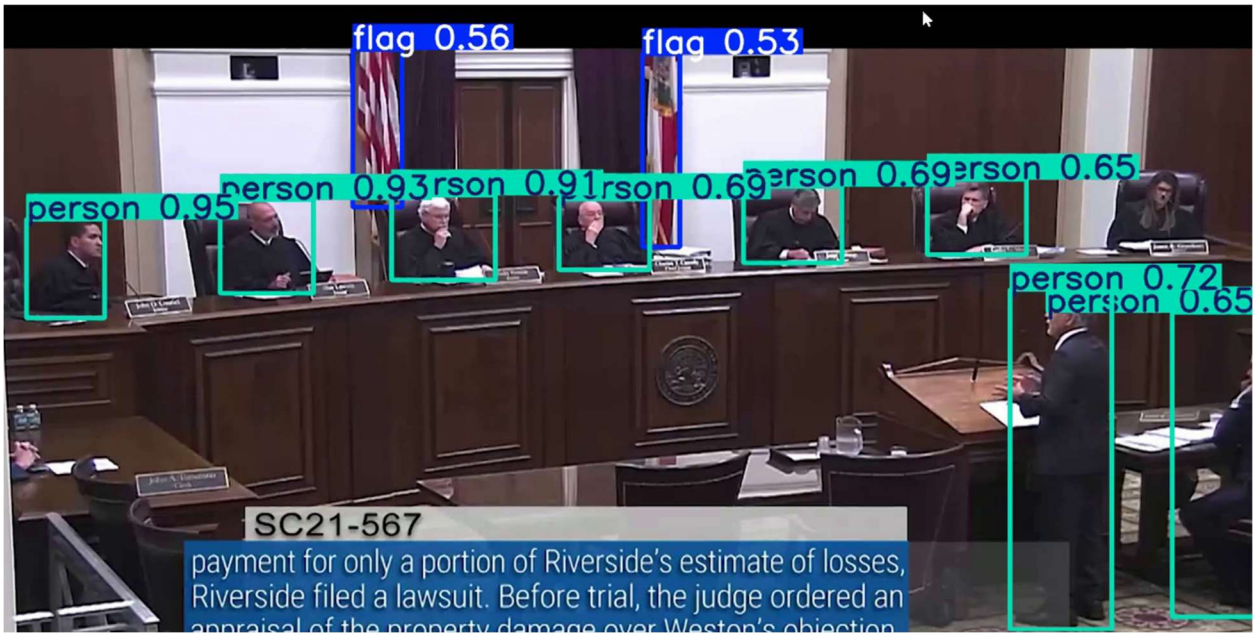
2.4 Testing and Detection

The trained model was tested on both images and video. Image testing confirmed that objects were correctly identified.





Video testing showed consistent detection across frames. Video detection example:



2.5 CV Results

The model performed well overall in detecting most courtroom objects. It showed high accuracy when identifying people and flags, which were consistently detected across both images

and video frames. The detection of the judge's bench was moderately accurate, with some variation depending on the angle and visibility in the scene. Microphone detection was less consistent compared to the other classes, likely due to their smaller size and variability in appearance. Despite these challenges, the model demonstrated strong overall performance in courtroom environments.

2.6 CV Limitations

The computer vision component had several limitations that impact its performance. One limitation is the relatively small dataset size, which restricts the model's ability to generalize to a wider range of courtroom scenarios. There is also some overlap between classes, which can occasionally lead to misclassification. Additionally, microphone detection proved to be less reliable, likely due to differences in shape, size, and positioning across images. These limitations highlight areas where further improvements could enhance the model's accuracy and robustness.

3. NLP Component

3.1 Overview

The NLP portion of the project focuses on analyzing the video's audio using transcription and text processing techniques.

3.2 Transcription

Audio from the video was transcribed using OpenAI's Whisper model. The model converted spoken dialogue into text with reasonable accuracy.

```python
import re

clean_text = re.sub(r'\s+', ' ', transcript).strip()

print("Cleaned transcript preview:")
print(clean_text[:1000])
```

```
Cleaned transcript preview:
denied all coverage, then there's no appraisal that's appropriate at this time. But if we're just being factually honest, there's been no eff
```

```python
# Split transcript into sentences
sentences = re.split(r'(?<=[.!?]) +', clean_text)

# Keep the first few meaningful sentences as a basic summary
summary = " ".join(sentences[:5])

print("SUMMARY:")
print(summary)
```

```
SUMMARY:
denied all coverage, then there's no appraisal that's appropriate at this time. But if we're just being factually honest, there's been no eff
```

3.3 Text Processing

After transcription, the text was processed to:

- Remove unnecessary spacing
- Improve readability
- Prepare for analysis

3.4 Summary Generation

A simple summarization approach was used by selecting the most relevant sentences from the transcript.

3.5 Keyword Extraction

Keywords were extracted using frequency-based analysis.

```
        "please", "thank", "good", "morning"
    }

    filtered_words = [word for word in words if word not in stopwords]

    keywords = Counter(filtered_words).most_common(15)

    print("KEYWORDS:")
    for word, count in keywords:
        print(f"{word}: {count}")
```

```
KEYWORDS:
coverage: 6
back: 4
waiver: 4
denied: 3
claw: 3
well: 3
appraisal: 2
advanced: 2
think: 2
eventually: 2
where: 2
accurate: 2
didn: 2
knew: 2
hurricane: 2
```

3.6 Chatbot Implementation

A transcript-based chatbot was developed to answer questions about the video. Instead of generating new answers, the chatbot retrieves the most relevant portion of the transcript based on the question.

```
questions = [
    "What is the issue with coverage?",
    "What does the speaker say about appraisal?",
    "Was coverage denied?",
    "What legal issue appears in the transcript?"
]

for q in questions:
    print("\nQ:", q)
    print("A:", courtroom_chatbot(q))
```

```
Q: What is the issue with coverage?
A: So I believe that it is accurate, it's accurate to say both. Okay, that we initially provided that, those funds, when we didn't know the a

Q: What does the speaker say about appraisal?
A: denied all coverage, then there's no appraisal that's appropriate at this time. But if we're just being factually honest, there's been no

Q: Was coverage denied?
A: denied all coverage, then there's no appraisal that's appropriate at this time. But if we're just being factually honest, there's been no

Q: What legal issue appears in the transcript?
A: Well, there couldn't be, there couldn't be a waiver based on something you did before. That's right. They're misconduct. They're alleged m
```

3.7 NLP Results

The NLP component of the system performed effectively in analyzing courtroom audio. It successfully transcribed spoken dialogue into text using the Whisper model, capturing key portions of legal discussions. From this transcription, the system was able to identify important legal terms through keyword extraction and generate basic summaries. Additionally, the chatbot component was able to answer simple questions by retrieving relevant parts of the transcript. Overall, the NLP system demonstrated the ability to extract meaningful information from courtroom audio.

3.8 NLP Limitations

Despite its functionality, the NLP component has several limitations. Its performance is highly dependent on the length and quality of the transcript, meaning shorter or unclear audio results in weaker analysis. The system also has limited contextual understanding, as it relies on surface-level text processing rather than deeper semantic reasoning. Furthermore, the chatbot is retrieval-based rather than generative, meaning it can only return existing portions of the transcript rather than create more advanced or synthesized responses.

4. Integration of CV and NLP

Although the computer vision and natural language processing components were developed separately, they were ultimately combined into a single system. The final pipeline begins with a video input, which is processed by the computer vision model to detect objects within the scene. At the same time, the audio is transcribed using the NLP component. The resulting transcript is then processed to generate summaries, extract keywords, and support chatbot-based question answering. This integrated pipeline allows the system to analyze both visual and spoken information, providing a more complete understanding of courtroom activity.

5. Results and Discussion

The integration of computer vision and natural language processing allowed the system to analyze both visual and spoken information from courtroom videos in a cohesive manner. The computer vision component demonstrated strong performance, particularly in detecting people and flags with high accuracy across both images and video frames. Detection of the judge's bench was generally reliable, although performance varied depending on camera angles and visibility. In contrast, microphone detection was less consistent, highlighting the challenges associated with smaller and less distinct objects. Overall, the CV model showed effective performance in identifying key courtroom elements.

The NLP component successfully extracted meaningful information from courtroom audio through transcription, keyword extraction, summarization, and basic question answering. The Whisper model produced accurate transcripts that enabled further analysis; however, the effectiveness of the NLP system was dependent on the length and clarity of the transcript. Shorter videos resulted in limited contextual information, which reduced the chatbot's ability to provide more detailed responses. Together, the CV and NLP components demonstrate the value of a multimodal approach, combining spatial and contextual understanding to produce a more comprehensive analysis of courtroom activity.

6. Hyperparameter Analysis

Parameter:	Value:	Effect:
Confidence	0.5	Balanced detections
IoU	0.4	Reduced overlap
Epochs	30	Improved accuracy

Adjusting these parameters affected detection performance and model stability.

7. Limitations

This project has several limitations that affect its overall performance and scope. First, the system does not operate in real time, as all processing is done after the video is uploaded and analyzed. Additionally, the dataset used for training was relatively small, which may limit the model's ability to generalize to more diverse courtroom environments. The NLP component is also constrained by the length of the transcripts, as shorter videos provide less information for meaningful analysis. Finally, the chatbot implementation is relatively basic, relying on retrieval-based logic rather than more advanced generative methods, which limits its ability to provide deeper contextual answers.

8. Future Improvements

There are several ways this project could be improved in the future. Expanding the dataset with more images and varied courtroom scenarios would likely improve detection accuracy. Implementing real-time processing would allow the system to analyze live video streams instead of pre-recorded footage. Adding speaker classification could help distinguish between roles such as judges, attorneys, and witnesses. The NLP component could also be enhanced by incorporating more advanced language models to provide more accurate and context-aware responses. Finally, developing a full user interface would make the system more accessible and interactive, allowing users to upload videos and receive analysis in a more streamlined way.

9. Conclusion

This project presented the development of a Courtroom Scene Analyzer that integrates both Computer Vision and Natural Language Processing to analyze courtroom video data. The computer vision component, built using a YOLOv8 model, successfully detected key courtroom objects such as people, flags, the judge's bench, and microphones. A custom dataset was created and labeled using RoboFlow, and the model demonstrated strong performance, particularly for larger and more consistent objects like people and flags. Although detection of smaller objects such as microphones was less reliable, the overall results showed that the model was capable of generalizing to real courtroom footage.

The natural language processing component utilized the Whisper model to transcribe courtroom audio into text, which was then processed to generate summaries, extract keywords, and support a transcript-based chatbot. While the NLP system effectively captured and analyzed key portions of dialogue, its performance was dependent on transcript length and clarity. Despite these limitations, the integration of both components resulted in a functional multimodal system capable of analyzing both visual and spoken information. Overall, this project demonstrates the potential of combining computer vision and NLP techniques to create more comprehensive and intelligent systems for real-world applications such as legal analysis.